3.2 Find the Laplace transform of the following time functions:

(a) $f(t) = 1 + 2t$

(b) $f(t) = 3 + 7t + t^2 + \delta(t)$

(c) $f(t) = e^{-t} + 2e^{-2t} + te^{-3t}$

(d) $f(t) = (t+1)^2$

(e) $f(t) = \sinh t$

3.3 Find the Laplace transform of the following time functions:

(a) $f(t) = 3\cos 6t$

(b) $f(t) = \sin 2t + 2\cos 2t + e^{-t}\sin 2t$

(c) $f(t) = t^2 + e^{-2t}\sin 3t$

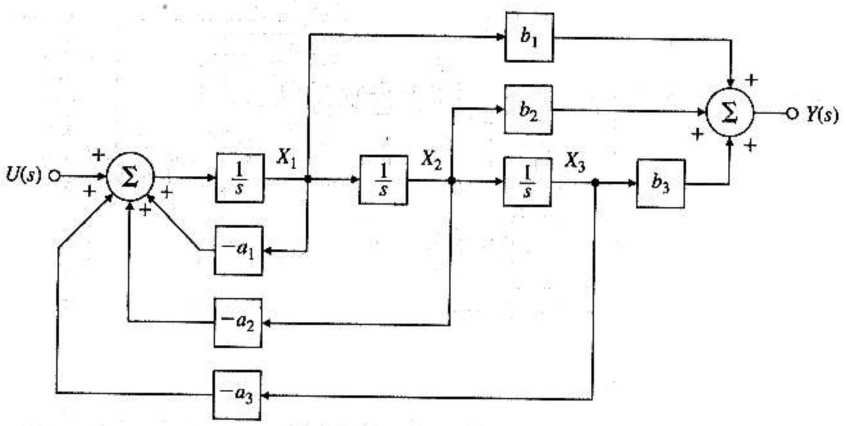
3.7 Find the time function corresponding to each of the following Laplace transforms using partial-fraction expansions:

(a) $F(s) = \frac{2}{s(s+2)}$

(b) $F(s) = \frac{10}{s(s+1)(s+10)}$ (c) $F(s) = \frac{3s+2}{s^2+4s+20}$

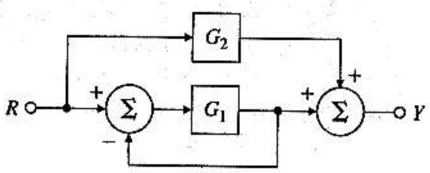
3.19 Consider the block diagram shown in Fig. 3.52. Note that a_i and b_i are constants. Compute the transfer function for this system. This special structure is called the "control canonical form" and will be discussed further in Chapter 7.

Figure 3.52
Block diagram for Problem 3.19



3.20 Find the transfer functions for the block diagrams in Fig. 3.53.

Figure 3.53
Block diagrams for Problem 3.20



(a)

3.21 Find the transfer functions for the block diagrams in Fig. 3.54, using the ideas of block diagram simplification. The special structure in Fig. 3.54(b) is called the "observer canonical form" and will be discussed in Chapter 7.

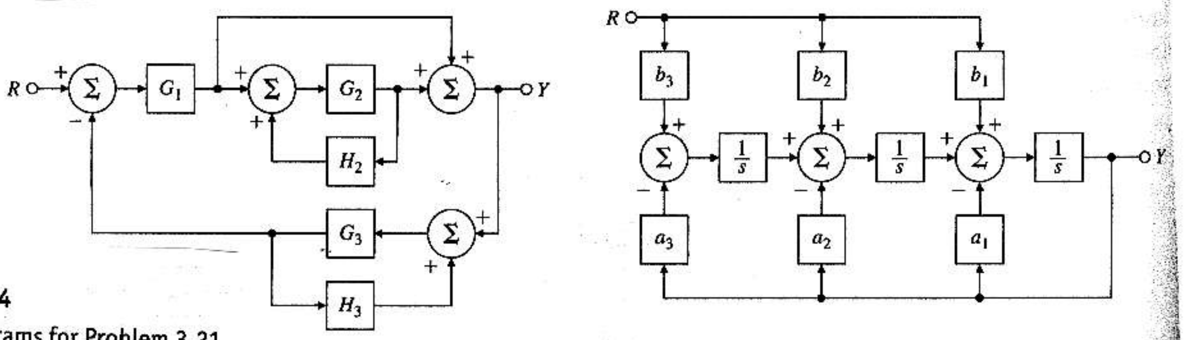


Figure 3.54
Block diagrams for Problem 3.21

(a) (b)